Second Calls and Mutual Aid

```
# load in packages and data
library(tidyverse)
library(lubridate)
second_aid <- read_csv("second_shift_aid.csv")

# create seasons col and levels for days and seasons for plots
days <- c("Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday")
seasons <- c("Spring", "Summer", "Fall", "Winter")
second_aid <- second_aid |>
  mutate(`Incident Date` = mdy(`Incident Date`),
         `Incident Day Name` = factor(`Incident Day Name`, levels = days),
         Season = case_when(month(`Incident Date`) %in% c(12, 01, 02) ~ "Winter",
                            month(`Incident Date`) %in% c(03, 04, 05) ~ "Spring",
                            month(`Incident Date`) %in% c(06, 07, 08) ~ "Summer",
                            month(`Incident Date`) %in% c(09, 10, 11) ~ "Fall")) |>
  mutate(Season = factor(Season, levels = seasons))
```

```
# table of capability of unit values
second_aid |>
  count(`Capability of Unit (eResponse.07)`)
```

```
# A tibble: 11 x 2
   `Capability of Unit (eResponse.07)`     n
   <chr>                                <int>
 1 Ambulance (ALS)                       5036
 2 Ambulance (BLS)                        828
 3 Ambulance (Critical Care)                1
 4 Call Turn Over / Mutual Aid            147
 5 Critical Care Ambulance                  2
 6 First Response                          25
 7 First Response (ALS)                   169
```

```
 8 First Response (BLS)                    25
 9 Intercept                                2
10 Intercept (ALS)                          4
11 <NA>                                     7
```

```
# number of mutual aid incidents
mutual_aid_count <- second_aid |>
  filter(`Capability of Unit (eResponse.07)` == "Call Turn Over / Mutual Aid") |>
  nrow()

# table of ems shift values
second_aid |>
  count(`EMS Shift (itResponse.005)`)
```

```
# A tibble: 9 x 2
  `EMS Shift (itResponse.005)`                n
  <chr>                                   <int>
1 911 Day                                   821
2 911 Night                                 426
3 A                                        2516
4 B                                        1545
5 Event/Standby                              29
6 No Shift (Second Call)                      6
7 No Shift (Second Call/Call-In Transfer)    49
8 Transfer Shift                            262
9 <NA>                                      592
```

```
# number of second call incidents
second_call_count <- second_aid |>
  filter(`EMS Shift (itResponse.005)` %in%
           c("No Shift (Second Call)", "No Shift (Second Call/Call-In Transfer")) |>
  nrow()
```

The total number of incidents not taken by the scheduled EMS crew from 2021-Present is 153.

```
# explore mysterious A and B entries
check <- second_aid |>
  filter(`EMS Shift (itResponse.005)` == "A" | `EMS Shift (itResponse.005)` == "B")

check |>
  head(10)
```

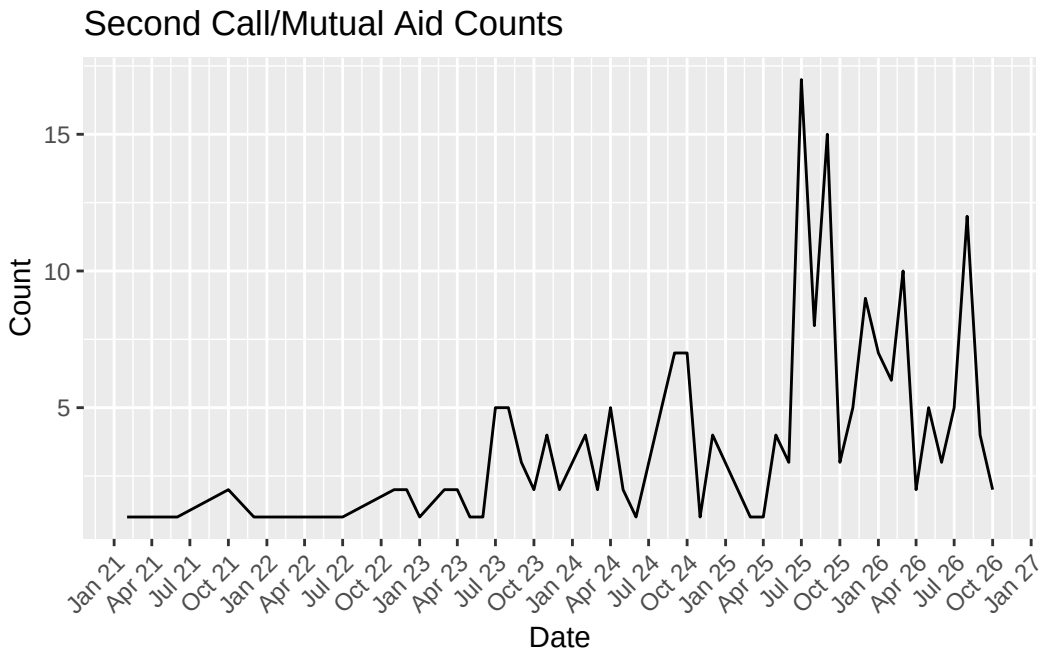
```
# A tibble: 10 x 7
   `Incident Date` `Incident Actual Time` `Incident Day Name`
   <date>          <time>                 <fct>
 1 2021-01-01      17:56:00               Friday
 2 2021-01-03      19:03:00               Sunday
 3 2021-01-04      02:02:00               Monday
 4 2021-01-04      11:20:00               Monday
 5 2021-01-04      20:57:00               Monday
 6 2021-01-05      20:12:00               Tuesday
 7 2021-01-05      10:11:19               Tuesday
 8 2021-01-05      09:22:00               Tuesday
 9 2021-01-06      09:25:00               Wednesday
10 2021-01-06      12:20:00               Wednesday
# i 4 more variables: `EMS Shift (itResponse.005)` <chr>,
#   `Capability of Unit (eResponse.07)` <chr>,
#   `Type of Call (eResponse.05)` <chr>, Season <fct>
```

```
most_recent_use <- check |>
  pull(`Incident Date`) |>
  max()
```

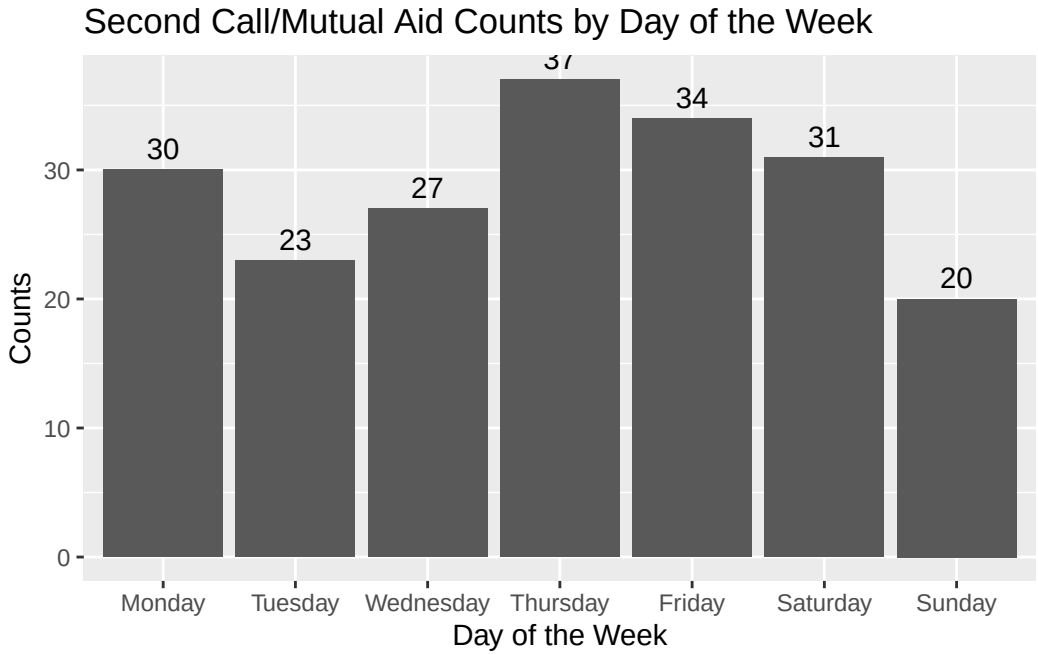
We were wondering what EMS Shift "A" and "B" were refering to. The last use of them was 2025-07-08.

```
# filter to only contain second shift and mutual aid
second_aid_filter <- second_aid |>
  filter(`EMS Shift (itResponse.005)` == "No Shift (Second Call/Call-In Transfer)" |
           `EMS Shift (itResponse.005)` == "No Shift (Second Call)" |
           `Capability of Unit (eResponse.07)` == "Call Turn Over / Mutual Aid")

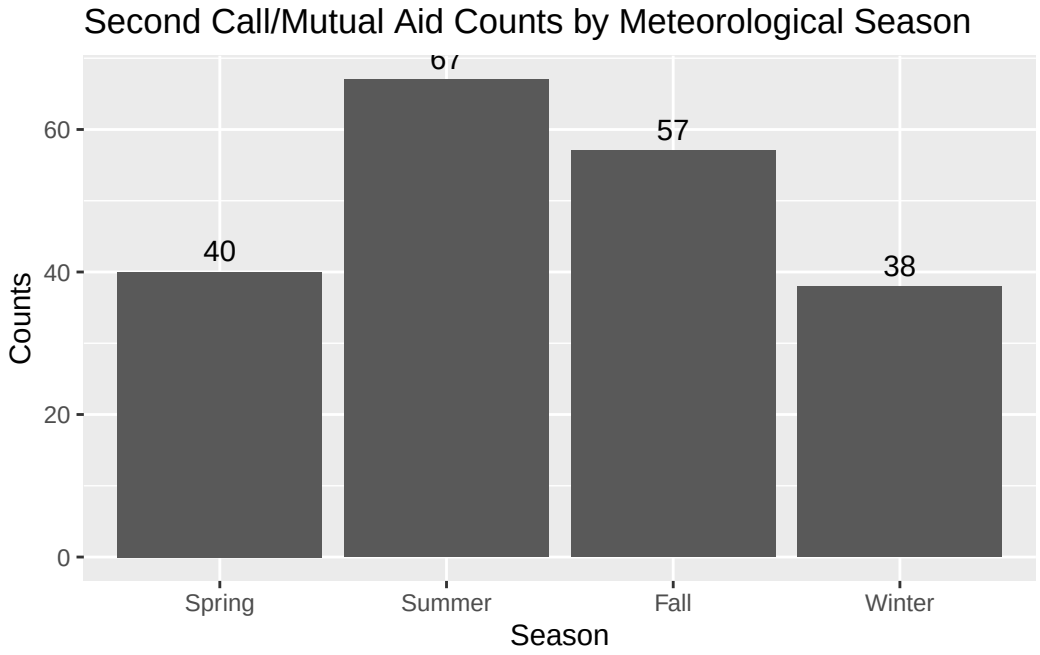
# full time series
second_aid_filter |>
  count(period = floor_date(`Incident Date`, "month")) |>
  ggplot() +
  geom_line(aes(x = period, y = n)) +
  scale_x_date(date_breaks = "3 month", date_labels = "%b %y") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(
    x = "Date",
    y = "Count",
    title = "Second Call/Mutual Aid Counts"
  )
```



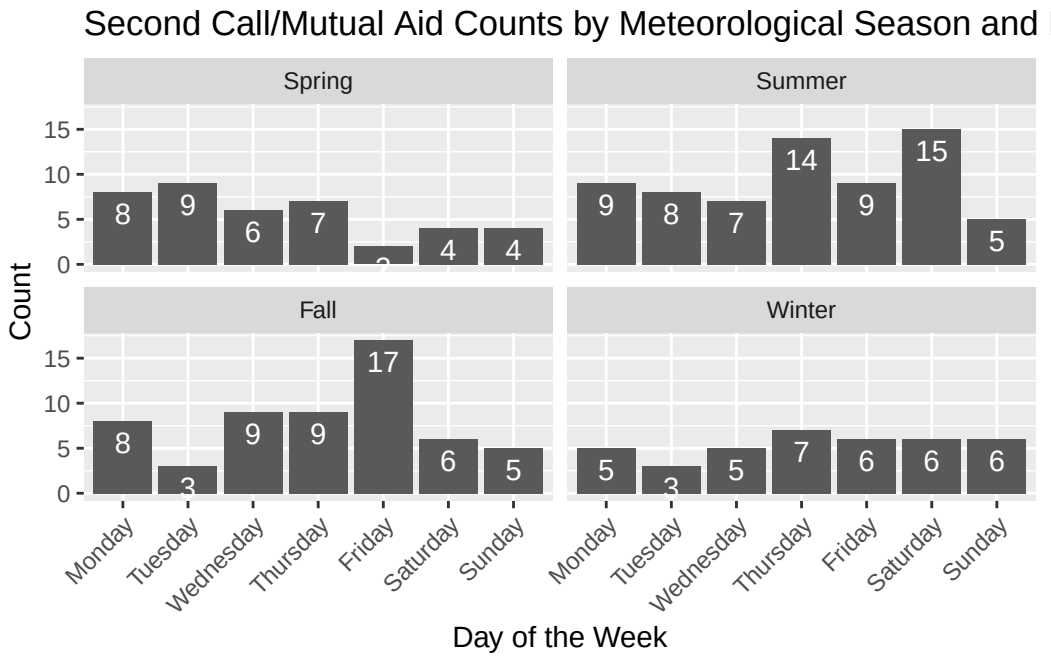

```
# bar dotw
second_aid_filter |>
  ggplot() +
  geom_bar(aes(x = `Incident Day Name`)) +
  geom_text(aes(x = `Incident Day Name`, label = after_stat(count)),
            stat = "count", vjust = -0.5) +
  labs(
    x = "Day of the Week",
    y = "Counts",
    title = "Second Call/Mutual Aid Counts by Day of the Week"
  )
```



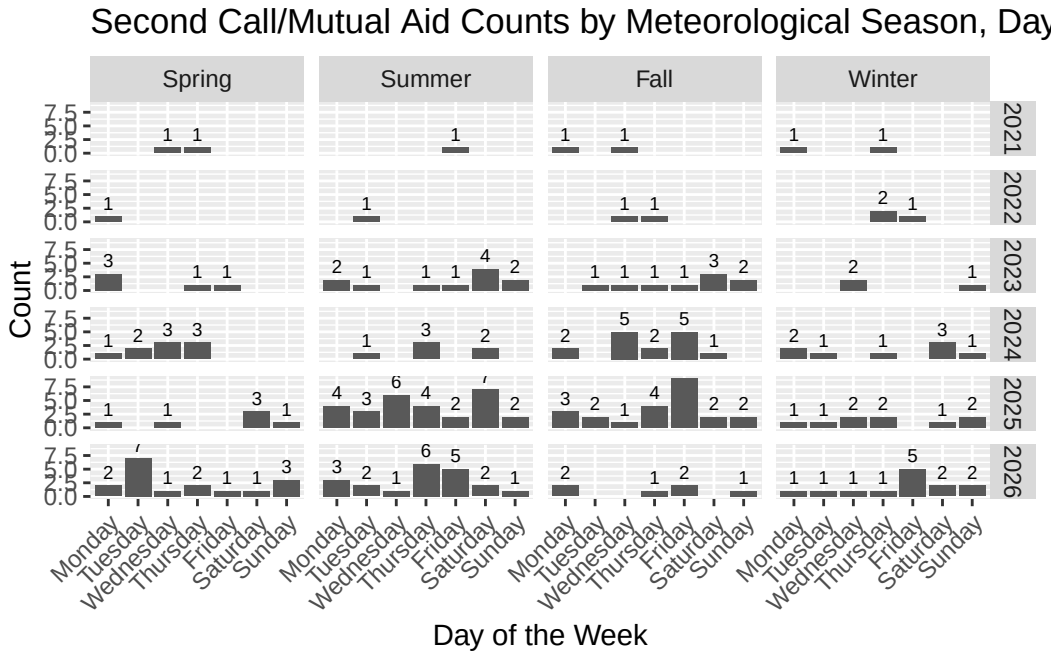

```
# bar season
second_aid_filter |>
  ggplot() +
  geom_bar(aes(x = Season)) +
  geom_text(aes(x = Season, label = after_stat(count)),
            stat = "count", vjust = -0.5) +
  labs(
    x = "Season",
    y = "Counts",
    title = "Second Call/Mutual Aid Counts by Meteorological Season"
  )
```




```
# bar dotw and season
second_aid_filter |>
  ggplot() +
  geom_bar(aes(x = `Incident Day Name`)) +
  facet_wrap(~Season) +
  geom_text(aes(x = `Incident Day Name`, label = after_stat(count)),
            stat = "count", vjust = 1.5, color = "white") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(
    x = "Day of the Week",
    y = "Count",
    title = "Second Call/Mutual Aid Counts by Meteorological Season and Day of the Week"
  )
```

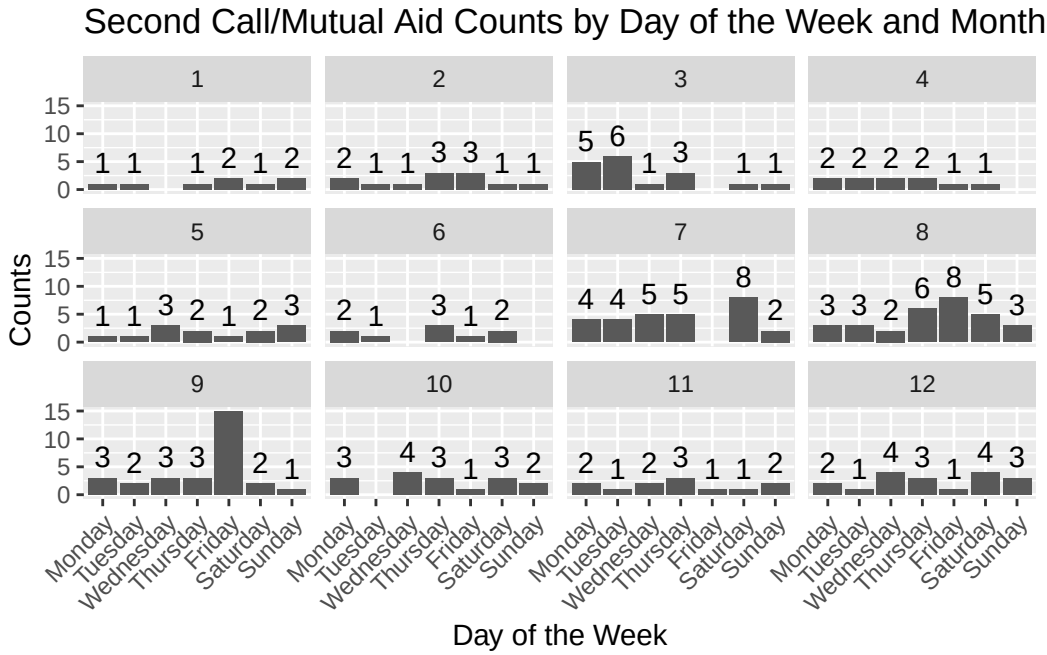



```
# bar dotw, season, and year
second_aid_filter |>
  ggplot() +
  geom_bar(aes(x = `Incident Day Name`)) +
  facet_grid(year(`Incident Date`) ~ Season) +
  geom_text(aes(x = `Incident Day Name`, label = after_stat(count)),
            stat = "count", vjust = -0.5, size = 2.5) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(
    x = "Day of the Week",
    y = "Count",
    title = "Second Call/Mutual Aid Counts by Meteorological Season, Day of the Week, and Yea
  )
```

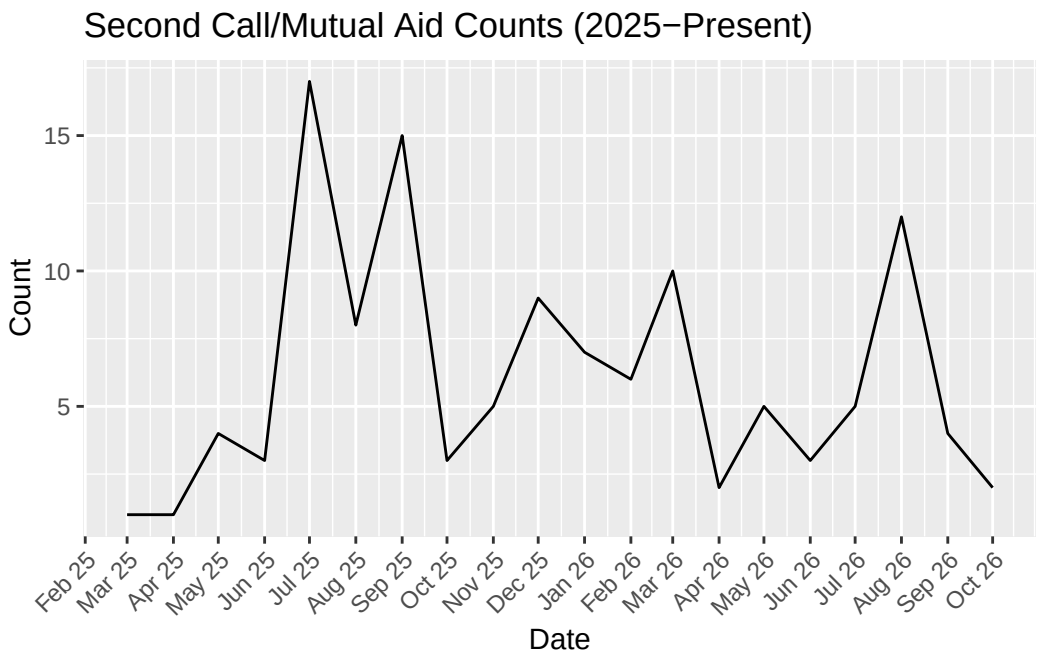



```
## 13 weeks in a season
```

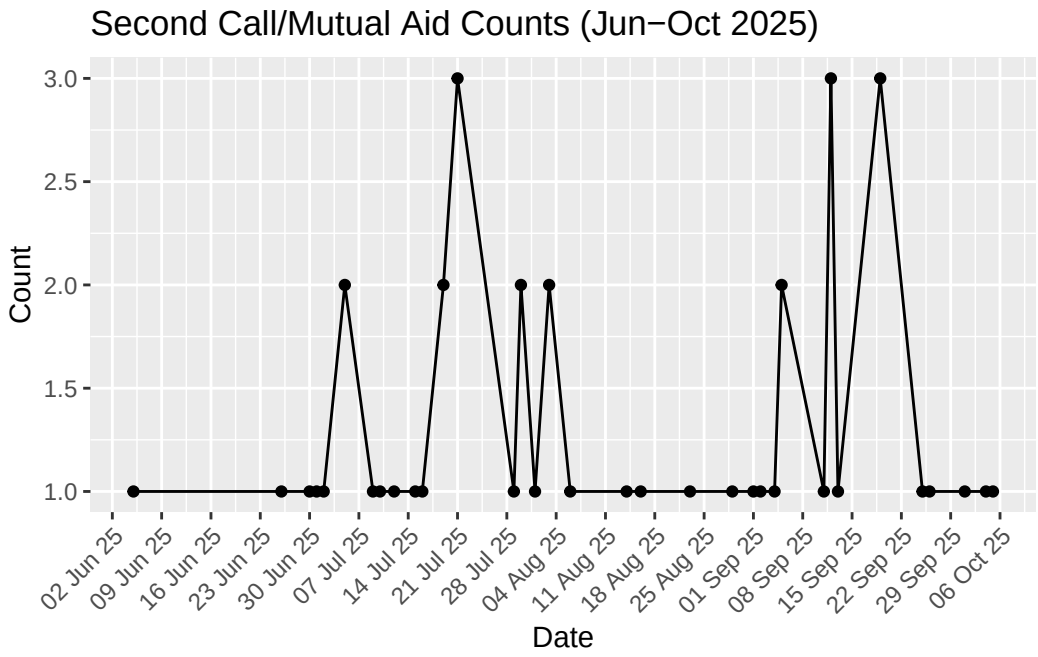
```
# bar month
second_aid_filter |>
  ggplot() +
  geom_bar(aes(x = `Incident Day Name`)) +
  facet_wrap(~month(`Incident Date`)) +
  geom_text(aes(x = `Incident Day Name`, label = after_stat(count)),
            stat = "count", vjust = -0.5) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(
    x = "Day of the Week",
    y = "Counts",
    title = "Second Call/Mutual Aid Counts by Day of the Week and Month"
  )
```




```r
# 2025-Present
second_aid_filter |>
  filter(year(`Incident Date`) >= 2025) |>
  count(period = floor_date(`Incident Date`, "month")) |>
  ggplot() +
  geom_line(aes(x = period, y = n)) +
  scale_x_date(date_breaks = "1 month", date_labels = "%b %y") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(
    x = "Date",
    y = "Count",
    title = "Second Call/Mutual Aid Counts (2025-Present)"
  )
```




```
# Jun 25 - Oct 25
second_aid_filter |>
  filter(year(`Incident Date`) == 2025 &
           month(`Incident Date`) > 05 &
           month(`Incident Date`) < 11) |>
  count(period = floor_date(`Incident Date`, "day")) |>
  ggplot(aes(x = period, y = n)) +
  geom_line() +
  geom_point() +
  scale_x_date(date_breaks = "1 week", date_labels = "%d %b %y") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(
    x = "Date",
    y = "Count",
    title = "Second Call/Mutual Aid Counts (Jun-Oct 2025)"
  )
```



Next steps would be to look at the max second calls in one day and see if we can determine any reasons as to why (weather, time of year, etc…)